

KPI Definitions and Business Metrics Handbook

****Document Type:**** Analytics Reference & Governance

****Applies To:**** AI Business Analyst, Analytics Teams, Management Reporting

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****Primary Demonstration Period:**** Q3 2011

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1. Purpose

This handbook establishes authoritative definitions for all key performance indicators (KPIs) and business metrics used by NovaMart. Its primary purpose is to ensure consistent, accurate, and comparable analysis across all reporting, dashboards, and AI-driven insights. The AI Business Analyst must apply these definitions when interpreting transaction data, and any deviation must be explicitly documented and approved by the Analytics Governance Committee.

This document does not contain KPI values, financial results, or performance data. It is a definitions and governance reference only. All metrics are defined in a manner that is directly measurable from the base transaction dataset, unless otherwise stated.

2. KPI Governance

- **Single Source of Truth:** All KPIs are derived from the cleaned transaction dataset. No external data sources are used unless explicitly integrated and documented.
- **Consistency:** Definitions are stable over time. Any change requires a version update and impact analysis.
- **Transparency:** Every metric must be traceable to its underlying calculation logic.
- **Data Cleaning Policy:** Cancelled transactions are excluded from all revenue and order metrics. Returned transactions (if present as negative quantities) are netted against revenue in the period they occur. Orders with missing or invalid Invoice IDs are discarded.
- **Confidence Levels:** Metrics are reported with a confidence indicator (High, Medium, Low) based on data completeness, sample size, and stability. The AI must not present low-confidence metrics as definitive.

3. Core Financial Metrics

3.1 Revenue

Definition: Revenue is the total monetary value of completed sales transactions.

Formula: $\text{Revenue} = \text{Quantity} \times \text{Unit Price}$

Handling of Cancellations and Returns:

- Cancelled orders (flagged as cancelled or with zero/negative net quantity) are excluded entirely.
- Returns, if present as separate negative-quantity transactions, are subtracted from gross revenue in the period they are recorded. This yields net revenue.
- No accounting-specific revenue recognition rules (e.g., deferred revenue, accrual adjustments) are applied, as they cannot be supported by the available transaction data.

3.2 Order Count

****Definition:**** An order is a unique purchase event identified by a single Invoice/Order ID.

****Formula:**** Count of distinct Invoice IDs for completed (non-cancelled) transactions.

****Treatment of Cancelled Orders:**** Cancelled orders are not counted. If an Invoice ID contains only cancelled line items, it is excluded.

3.3 Average Order Value (AOV)

****Definition:**** The average monetary value of a completed order.

****Formula:**** Average Order Value = Total Revenue / Number of Completed Orders

3.4 Revenue Growth

****Definition:**** The percentage change in revenue between two comparable periods.

****Formula:****

Revenue Growth % = ((Current Period Revenue – Previous Period Revenue) / Previous Period Revenue) × 100

****Note:**** Previous period must be of equal length and comparable seasonality. For Q3 2011, the previous period is Q3 2010 or Q2 2011, depending on the analysis requirement.

3.5 Revenue Contribution

****Definition:**** The share of total revenue attributable to a specific segment (country, region, product, category, customer segment).

****Formula:**** Revenue Contribution % = (Segment Revenue / Total Revenue) × 100

****Application:**** Used for country, region, product, category, and customer segment analysis.

3.6 Discount Rate

****Definition:**** The percentage of gross revenue that is reduced by discounts.

****Formula (if discount data available):**** Discount Rate % = (Total Discount Amount / Total Gross Revenue) × 100

****Current Dataset Status:**** ****Not directly measurable from the base transaction dataset without additional data.**** The base transaction dataset does not contain discount fields. Any discount analysis requires integration with a separate promotions or pricing table.

3.7 Profit Margin

****Definition:**** The percentage of revenue remaining after deducting the cost of goods sold (COGS).

****Formula:**** $\text{Profit Margin \%} = ((\text{Revenue} - \text{COGS}) / \text{Revenue}) \times 100$

****Current Dataset Status:**** ****Cannot be calculated from transaction data alone.**** The base transaction dataset does NOT contain actual cost of goods sold. True profit margin requires a separate cost table. Do not invent cost figures or estimate margins without approved cost data.

4. Customer Metrics

4.1 Customer Count

- ****Unique Customers:**** Count of distinct Customer IDs in the dataset (all time or within a period).
- ****Active Customers:**** Count of distinct Customer IDs with at least one completed purchase in the specified period.
- ****Repeat Customers:**** Count of distinct Customer IDs with more than one completed purchase. This can be measured within a single period (e.g., two orders in Q3 2011) or on a lifetime basis (at least two orders ever). The context must be stated.

4.2 Repeat Customer Rate

****Definition:**** The proportion of active customers in a period who are repeat customers.

****Formula:****

Repeat Customer Rate % = (Number of Customers with >1 Completed Purchase in Period / Total Active Customers in Period) × 100

****Alternative (Lifetime):**** (Number of Customers with >1 Lifetime Purchase / Total Unique Customers) × 100. Specify which definition is used.

4.3 Customer Revenue

****Definition:**** Total revenue associated with a specific customer.

****Formula:**** Sum of Revenue for all completed orders belonging to that Customer ID.

4.4 Customer Activity

- ****Purchase Frequency:**** Average number of completed orders per active customer in a period. Formula: Total Completed Orders / Total Active Customers.

- ****Recency:**** Number of days since the customer's last completed purchase, relative to the analysis date.

- ****Order Count:**** Total number of completed orders for a customer (period or lifetime).

5. Product Metrics

5.1 Units Sold

****Definition:**** Total quantity of a product sold in completed transactions.

****Formula:**** Sum of Quantity for the product across all completed orders.

5.2 Product Revenue

****Definition:**** Total revenue generated by a product.

****Formula:**** Sum of (Quantity × Unit Price) for the product.

5.3 Product Revenue Growth

****Definition:**** Percentage change in product revenue between two periods.

****Formula:**** Same as Revenue Growth, applied to a specific product.

5.4 Product Contribution

****Definition:**** The share of total revenue attributable to a specific product.

****Formula:**** $\text{Product Contribution \%} = (\text{Product Revenue} / \text{Total Revenue}) \times 100$

6. Regional Metrics

6.1 Growth by Country

****Definition:**** Revenue growth for a specific country between two periods.

****Formula:****

$\text{Country Growth \%} = ((\text{Current Period Country Revenue} - \text{Previous Period Country Revenue}) / \text{Previous Period Country Revenue}) \times 100$

6.2 Growth by Region

****Definition:**** Revenue growth for a reporting region (UK, Continental Europe, APAC, Other International).

****Formula:**** Same as above, applied to region-level revenue.

6.3 Regional Revenue Contribution

****Definition:**** Share of total revenue by region.

****Formula:**** $(\text{Region Revenue} / \text{Total Revenue}) \times 100$

7. Analytical Metrics

7.1 Anomaly

****Definition:**** An unusual movement in a metric relative to recent historical behavior. This is typically identified as a value that falls outside an expected range (e.g., more than two standard deviations from a moving average, or a sudden spike/drop not explained by seasonality).

****Important:**** Anomaly detection is an analytical signal, not proof of a cause. It indicates that further investigation is warranted.

7.2 Growth by Product

****Definition:**** Revenue growth for a specific product between two periods.

****Formula:****
$$\frac{(\text{Current Period Product Revenue} - \text{Previous Period Product Revenue})}{\text{Previous Period Product Revenue}} \times 100$$

7.3 Growth by Country

****Definition:**** Revenue growth for a specific country between two periods.

****Formula:****
$$\frac{(\text{Current Period Country Revenue} - \text{Previous Period Country Revenue})}{\text{Previous Period Country Revenue}} \times 100$$

8. Interpretation Rules

8.1 Fact vs. Interpretation

- **Fact:** A statement directly derived from measurable data. Example: “Revenue declined 15% in Q3 2011 compared to Q3 2010.”
- **Interpretation:** An explanation, opinion, or causal claim. Example: “Distributor problems caused the decline.” Interpretations require additional evidence and must be labeled as such.

8.2 Correlation vs. Causation

Correlation between two metrics does not imply that one causes the other. A causal claim requires:

- Temporal precedence (cause before effect)
- Mechanism or plausible explanation
- Elimination of confounding variables
- Supporting evidence beyond the correlation itself.

8.3 High Confidence

A KPI has high confidence when:

- Data is complete with no missing critical fields.
- Sample size is sufficiently large (e.g., >30 observations for averages).
- The metric is stable across sub-periods.
- No known data quality issues exist.

8.4 Insufficient Data

Data is insufficient when:

- Key fields are missing or null.
- Sample size is too small to draw reliable conclusions.
- The period is incomplete or not comparable.
- External factors make the data unrepresentative.

In such cases, the AI must explicitly state that the metric cannot be reliably calculated.

9. Limitations

- **No Cost Data:** Profit margin cannot be calculated.
- **No Discount Data:** Discount rate cannot be calculated from the base dataset.
- **No Customer Demographics:** Segmentation is limited to behavioral definitions.
- **No Marketing Spend:** ROI and attribution cannot be measured.
- **Historical Period:** The dataset covers 2009–2011. Business conditions may differ from present day.
- **Cancellations and Returns:** Handling depends on how these are flagged in the dataset. Assumptions must be documented.

10. KPI Reference Table

Metric	Definition	Formula	Business Use	Limitations
Revenue	Total value of completed sales	Quantity × Unit Price (net of returns, excl. cancellations)	Core financial performance	No COGS, no discounts
Order Count	Number of unique completed orders	Count distinct Invoice IDs	Volume measurement	Cancelled orders excluded
Average Order Value	Average revenue per completed order	Total Revenue / Completed Orders	Pricing and promotion effectiveness	Sensitive to outliers

| Customer Count (Unique) | Distinct customers | Count distinct Customer IDs | Market reach | Does not indicate activity |

| Active Customers | Customers with ≥ 1 purchase in period | Count distinct Customer IDs with ≥ 1 completed order | Engagement measurement | Period-dependent |

| Repeat Customers | Customers with > 1 purchase | Count distinct Customer IDs with > 1 completed order | Loyalty indicator | Definition varies (period vs lifetime) |

| Repeat Customer Rate | % of active customers who are repeat | $(\text{Repeat Customers} / \text{Active Customers}) \times 100$ | Retention performance | Requires clear period definition |

| Revenue Growth % | Change in revenue between periods | $((\text{Current} - \text{Previous}) / \text{Previous}) \times 100$ | Trend analysis | Needs comparable periods |

| Revenue Contribution % | Share of total revenue by segment | $(\text{Segment Revenue} / \text{Total Revenue}) \times 100$ | Portfolio analysis | Segment definitions must be consistent |

| Units Sold | Quantity of product sold | Sum of Quantity | Product demand | Does not reflect revenue |

| Product Revenue | Revenue from a product | $\text{Sum}(\text{Quantity} \times \text{Unit Price})$ | Product performance | No cost data |

| Product Contribution % | Share of total revenue by product | $(\text{Product Revenue} / \text{Total Revenue}) \times 100$ | Portfolio optimization | Depends on total revenue accuracy |

| Customer Revenue | Total revenue per customer | Sum of Revenue for customer | CLV approximation | No cost-to-serve |

| Purchase Frequency | Orders per customer | $\text{Total Orders} / \text{Active Customers}$ | Engagement | Can be skewed by high-frequency buyers |

| Recency | Days since last purchase | $\text{Analysis Date} - \text{Last Purchase Date}$ | Churn risk | Requires reference date |

| Discount Rate | % of revenue discounted | $(\text{Discount Amount} / \text{Gross Revenue}) \times 100$ | Promotion effectiveness | **Not measurable from base dataset** |

| Profit Margin | % profit after COGS | $((\text{Revenue} - \text{COGS}) / \text{Revenue}) \times 100$ | Profitability | **Cannot be calculated without cost data** |

| Growth by Country | Revenue change by country | $((\text{Current Country Rev} - \text{Previous}) / \text{Previous}) \times 100$ | Regional performance | Needs comparable periods |

| Growth by Product | Revenue change by product | $((\text{Current Product Rev} - \text{Previous}) / \text{Previous}) \times 100$ | Product lifecycle | Small products may have volatile rates |

| Anomaly | Unusual movement vs. history | Statistically significant deviation | Early warning signal | Not proof of cause |
